

Internal code

PACLITAXEL SEACROSS 6 MG/ML CONCENTRATE FOR SOLUTION FOR INFUSION

QUALITATIVE AND QUANTITATIVE COMPOSITION

1 ml of concentrate for solution for infusion contains 6 mg paclitaxel.
Each vial of 5 ml contains 30 mg of paclitaxel.
Each vial of 16.7 ml contains 100 mg of paclitaxel.
Each vial of 50 ml contains 300 mg of paclitaxel.

PHARMACEUTICAL FORM

Concentrate for solution for infusion.
Before dilution:
A clear, colourless to slightly yellow viscous solution.
After dilution:
It's colorless and clear liquid.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Paclitaxel Seacross is indicated for the treatment of the following:
Ovarian Carcinoma
First-line therapy in combination with a platinum compound for the treatment of advanced metastatic carcinoma of the ovary.
Second-line therapy for the treatment of advanced metastatic carcinoma of the ovary.
Breast Carcinoma
Initial treatment of advanced or metastatic breast cancer in combination with trastuzumab, in patients who over-express HER-2 at a 3+ level as determined by immunohistochemistry and for whom an anthracycline is not suitable.
Second-line therapy for treatment of metastatic carcinoma of the breast, after failure of standard therapy.
Non-Small Cell Lung Carcinoma
First-line therapy in combination with a platinum compound or as a single agent for the treatment of non-small cell carcinoma of the lung in patients who are not candidates for potentially curative surgery and/or radiation therapy.

4.2 Dosage and Administration

Note: Contact of the undiluted concentrate with plasticized PVC equipment or devices used to prepare solutions for infusion is not recommended. In order to minimize patient exposure to the plasticizer DEHP [di-(2 ethylhexyl)phthalate], which may be leached from PVC infusion bags or sets, diluted Paclitaxel Seacross solutions must be stored in bottles (glass, polypropylene) or plastic bags (polypropylene, polyolefin) and administered through polyethylene-lined administration sets.

All patients must be premedicated prior to Paclitaxel Seacross administration to reduce the risk of severe hypersensitivity reactions. Such premedication may consist of dexamethasone 20mg orally (or its equivalent) approximately 12 and 6 hours before Paclitaxel Seacross or 20 mg IV approximately 30 to 60 min before Paclitaxel Seacross, diphenhydramine 50 mg I.V. (or its equivalent) 30 to 60 minutes prior to Paclitaxel Seacross and cimetidine (300 mg) or ranitidine (50 mg) I.V. 30 to 60 minutes prior to Paclitaxel Seacross.

Metastatic carcinoma of the ovary

Combination therapy: For previously untreated patients, the recommended dosing regimen, given every 3 weeks, is Paclitaxel Seacross administered intravenously over 3 hours at a dose of 175 mg/m² followed by a platinum compound.
Alternatively, a more myelosuppressive regimen of Paclitaxel Seacross may also be administered intravenously at a dose of 135 mg/m² over 24 hours followed by a platinum compound, every 3 weeks.

Single-agent therapy: In patients previously treated with chemotherapy the recommended regimen is 175 mg/m² administered intravenously over 3 hours every 3 weeks.

Breast Carcinoma

Combination therapy: When used in combination with trastuzumab, the recommended dose of Paclitaxel Seacross is 175 mg/m² administered intravenously over a period of 3 hours, with a 3-week interval between courses. Paclitaxel Seacross infusion may be started the day following the first dose of trastuzumab or immediately after the subsequent doses of trastuzumab if the preceding dose of trastuzumab was well tolerated.

Single-agent therapy: Paclitaxel Seacross 175 mg/m² administered intravenously over 3 hours every 3 weeks.

Non-small cell lung carcinoma

Combination therapy: For previously untreated patients, the recommended dosing regimen given with a 3-week interval between courses is, Paclitaxel Seacross 175 mg/m² administered intravenously over 3 hours followed by a platinum compound.

Alternatively, a more myelosuppressive regimen of Paclitaxel Seacross may be administered intravenously 135 mg/m² over 24 hours followed by a platinum compound, with a 3-week interval between courses.

Single-agent therapy: Paclitaxel Seacross 175 to 225 mg/m² administered intravenously over 3 hours every 3 weeks.

Repeat courses of Paclitaxel Seacross should not be administered to patients with solid tumors until the neutrophil count is at least 1,500 cells/mm³ and the platelet count is at least 100,000 cells/mm³ (<1000 cells/mm³ for patients with Kaposi's sarcoma). Patients who experience severe neutropenia (< 500 cells/mm³) or severe peripheral neuropathy should receive a dosage reduced by 20% for subsequent courses. The incidence of neurotoxicity and the severity of neutropenia increase with dose within a regimen.

Hepatic Impairment

Patients with hepatic impairment may be at increased risk of toxicity, particularly grade III-IV myelosuppression. Dose adjustment is recommended, as shown in Table below for both 3- and 24-hour infusions. Patients should be monitored closely for the development of profound myelosuppression. (see Pharmacokinetics; Hepatic Impairment.)

Table 1: Recommendations for Dosing in Patients with Hepatic Impairment

Degree of Hepatic Impairment			
Transaminase Levels		Bilirubin Levels	Recommended Paclitaxel Seacross Dose*
24-hour infusion			
< 2 x ULN	and	≤1.5mg/dL	135 mg/m ²
2-<10 x ULN	and	≤1.5mg/dL	100 mg/m ²
< 10 x ULN	and	1.6 – 7.5 mg/dL	50 mg/m ²
≥ 10 x ULN	or	>7.5 mg/dL	Not recommended
3-hour infusion			
< 10 x ULN	and	≤ 1.25mg/dL	175 mg/m ²
< 10 x ULN	and	1.26 – 2.0 x ULN	135 mg/m ²
< 10 x ULN	and	2.01 – 5.0 x ULN	90 mg/m ²
≥ 10 x ULN	or	>5.0 x ULN	Not recommended

*Dosage recommendations are for the first course of therapy; further dose reduction in subsequent courses should be based on individual tolerance.
ULN = upper limit of normal.

Preparation and Administration Precautions: Paclitaxel Seacross is a cytotoxic anticancer drug and caution should be exercised in handling Paclitaxel Seacross. The use of gloves is recommended. If Paclitaxel Seacross solution contacts the skin, wash the skin immediately and thoroughly with soap and water. If Paclitaxel Seacross contacts mucous membranes, the membranes should be flushed thoroughly with water. Following topical exposure, events have included tingling, burning and redness. Upon inhalation, dyspnea, chest pain, burning eyes, sore throat and nausea have been reported. Given the possibility of extravasation, it is advisable to closely monitor the infusion site for possible infiltration during drug administration (see Special warnings and precautions for use: Injection Site Reaction section).

Paclitaxel Seacross should be administered through an in-line filter with a microporous membrane not greater than 0.22 microns. Use of filter devices such as IVEX-2 which incorporate short inlet and outlet PVC-coated tubing has not resulted in significant leaching of DEHP.

Preparation for Intravenous Administration: Paclitaxel Seacross must be diluted prior to infusion to a final concentration of 0.3 to 1.2 mg/mL. Paclitaxel Seacross should be diluted in one of the following: 0.9% Sodium Chloride Injection, 5% Dextrose Injection and 5% Dextrose and 0.9% Sodium Chloride Injection.

Upon preparation, solutions may show haziness, which is attributed to the formulation vehicle. No significant losses in potency have been noted following delivery of the solution through intravenous tubing containing an in-line 0.22-micron filter.

Paclitaxel Seacross solutions should be prepared and stored in glass, polypropylene, or polyolefin containers. Non-PVC containing administration sets, such as those that are polyethylene-lined, should be used. (See Note.)

Diluted solutions are chemically and physically stable at 2°C to 8°C and at 25°C for 7 days when diluted in 5% dextrose solution, 5% dextrose and 0.9% sodium chloride solution and for 14 days when diluted in 0.9% sodium chloride injection.

From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and would normally not be longer than 24 hours at 2 to 8°C, unless dilution has taken place in controlled and validated aseptic conditions. After dilution the solution is for single use only.

Diluted solutions are chemically and physically stable for up to 27 hours at ambient temperature (approximately 25° C) and room lighting conditions; infusions should be completed within this timeframe. There have been rare reports of precipitation usually towards the end of a 24-hour infusion period. Excessive agitation, vibration or shaking may induce precipitation and should be avoided. Infusion sets should be flushed thoroughly with a compatible diluent before use.

Devices with spikes should not be used with vials of Paclitaxel Seacross since they can cause the stopper to collapse resulting in loss of sterile integrity of the Paclitaxel Seacross solution.

4.3 Contraindications

Paclitaxel Seacross is contraindicated in patients who have a history of severe hypersensitivity reactions to paclitaxel or polyoxyethylated castor oil.

Paclitaxel Seacross should not be administered to patients with solid tumors who have baseline neutrophil counts of < 1,500 cells/mm³ or patients with AIDS-related Kaposi's Sarcoma with baseline or subsequent neutrophil counts of <1,000 cells/mm³. (See DOSAGE AND ADMINISTRATION section.)

4.4 Special warnings and precautions for use

Paclitaxel Seacross should be administered under the supervision of a physician experienced in the use of cancer chemotherapeutic agents.

Paclitaxel Seacross should be administered as a diluted infusion. Patients must be treated with corticosteroids, antihistamines, and H2 antagonists before receiving Paclitaxel Seacross (See **DOSAGE AND ADMINISTRATION** section.)

Paclitaxel Seacross should be given before a platinum compound when it is given in combination with a platinum compound. (See Special warnings and precautions for use section.)

Anaphylaxis and severe hypersensitivity reactions

Anaphylaxis and severe hypersensitivity reactions have occurred commonly in patients receiving Paclitaxel Seacross. These reactions are probably histamine-mediated. Rare fatal reactions have occurred in patients despite pre-treatment. All patients should be pretreated with corticosteroids, diphenhydramine, and H2 antagonists. (See DOSAGE AND ADMINISTRATION section.) In case of a severe hypersensitivity reaction, Paclitaxel Seacross infusion should be discontinued immediately and the patient should not be retreated with Paclitaxel Seacross.

Hematologic Toxicity

Bone marrow suppression (primarily neutropenia) is dose and schedule dependent and is the principal dose-limiting toxicity within a regimen. Frequent monitoring of blood counts should be instituted during Paclitaxel Seacross treatment. Paclitaxel Seacross should not be administered to patients with baseline neutrophil counts of less than 1,500 cells/mm³ (<1,000 cells/mm³ for patients with Kaposi's Sarcoma). In case of severe neutropenia (< 500 cells/mm³) during a course of Paclitaxel Seacross, a 20% reduction in dose for subsequent courses of therapy is recommended. (See DOSAGE AND ADMINISTRATION sections.)

Cardiovascular Toxicity

Hypotension, hypertension and bradycardia have been observed during Paclitaxel Seacross administration; patients are usually asymptomatic and generally do not require treatment. In severe cases, Paclitaxel Seacross infusions may need to be interrupted or discontinued at the discretion of the treating physician. Frequent monitoring of vital signs, particularly during the first hour of Paclitaxel Seacross infusion is recommended. Continuous electrocardiographic monitoring is not required except for patients with serious conduction abnormalities. When Paclitaxel Seacross is used in combination with trastuzumab for treatment of metastatic breast cancer, monitoring of cardiac function is recommended. (See WARNINGS and ADVERSE EVENTS sections.)

Nervous System

The occurrence of peripheral neuropathy is frequent, but usually not severe. A dose reduction of 20% for subsequent courses of Paclitaxel Seacross is recommended for severe neuropathy. (See DOSAGE AND ADMINISTRATION and ADVERSE EVENTS sections.) Paclitaxel Seacross contains dehydrated ethanol. Consideration should be given to possible CNS and other effects of ethanol for all patients. Children may be more sensitive than adults to the effects of ethanol (see Pediatric Use section).

Injection Site Reaction

A specific treatment for extravasation reactions is unknown at this time. Given the possibility of extravasation, it is advisable to closely monitor the infusion site for possible infiltration during drug administration.

Hepatic Impairment

Patients with hepatic impairment may be at increased risk of toxicity, particularly grade III-IV myelosuppression. Dose adjustment is recommended. Patients should be monitored closely for the development of profound myelosuppression. (See PHARMACOLOGICAL PROPERTIES; Pharmacokinetics section.)

Pediatric Use: The safety and effectiveness of Paclitaxel Seacross in pediatric patients has not been established. There have been reports of central nervous system (CNS) toxicity (rarely associated with death) pediatric patients in which Paclitaxel Seacross was infused intravenously over 3 hours at doses ranging from 350 mg/m² to 420 mg/m². The toxicity is most likely attributable to the high dose of the ethanol component of the Paclitaxel Seacross vehicle given over a short infusion time. The use of concomitant antihistamines may intensify this effect. Although a direct effect of the paclitaxel itself cannot be discounted, the high doses used (over twice the recommended adult dosage) must be considered in assessing the safety of Paclitaxel Seacross for use in this population.

4.5 Interaction with other medicinal products and other forms of interaction

The recommended regimen of paclitaxel administration for the first-line chemotherapy of ovarian carcinoma is for paclitaxel to be given before cisplatin. When paclitaxel is given before cisplatin, the safety profile of paclitaxel is consistent with that reported for single-agent use. When paclitaxel was given after cisplatin, patients showed a more profound myelosuppression and an approximately 20% decrease in paclitaxel clearance. Patients treated with paclitaxel and cisplatin may have an increased risk of renal failure as compared to cisplatin alone in gynaecological cancers.

Since the elimination of doxorubicin and its active metabolites can be reduced when paclitaxel and doxorubicin are given closer in time, paclitaxel for initial treatment of metastatic breast cancer should be administered 24 hours after doxorubicin.

The metabolism of paclitaxel is catalysed, in part, by cytochrome P450 isoenzymes CYP2C8 and 3A4 (see section 5.2). Clinical studies have demonstrated that CYP2C8-mediated metabolism of paclitaxel, to 6α-hydroxypaclitaxel, is the major metabolic pathway in humans. Concurrent administration of ketoconazole, a known potent inhibitor of CYP3A4, does not inhibit the elimination of paclitaxel in patients; thus, both medicinal products may be administered together without dosage adjustment. Further data on the potential of drug interactions between paclitaxel and other CYP3A4 substrates/inhibitors are limited. Therefore, caution should be exercised when administering paclitaxel concomitantly with medicines known to inhibit (e.g. erythromycin, fluoxetine, gemfibrozil) or induce (e.g. rifampicin, carbamazepine, phenytoin, phenobarbital, efavirenz, nevirapine) either CYP2C8 or CYP3A4.

Paclitaxel clearance is not affected by cimetidine premedication.

Studies in KS patients, who were taking multiple concomitant medications, suggest that the systemic clearance of paclitaxel was significantly lower in the presence of nelfinavir and ritonavir, but not with indinavir. Insufficient information is available on interactions with other protease inhibitors. Consequently, paclitaxel should be administered with caution in patients receiving protease inhibitors as concomitant therapy.

4.6 Fertility, pregnancy and lactation

Pregnancy

There is no adequate information on the use of paclitaxel in pregnant women. Paclitaxel has been shown to be embryotoxic and foetotoxic in rabbits, and to decrease fertility in rats.

As with other cytotoxic drugs, paclitaxel may cause foetal harm, and is therefore contraindicated during pregnancy. Women should be advised to avoid becoming pregnant during therapy with paclitaxel, and to inform the treating physician immediately should this occur. Female and male patients of fertile age, and/or their partners should use contraceptions for at least 6 months after treatment with paclitaxel.

Breast-feeding

Paclitaxel is contraindicated during lactation. It is not known whether paclitaxel is excreted in human milk. Breastfeeding should be discontinued for the duration of therapy.

Fertility

Paclitaxel has been shown to reduce fertility in rats. Male patients should seek advice regarding cryoconservation of sperm prior to treatment with paclitaxel because of the possibility of infertility.

4.7 Effects on ability to drive and use machines

Since Paclitaxel Seacross contains ethanol, consideration should be given to the possibility of CNS and other effects. Consideration should be given to possible CNS effects of premedications given to reduce the risk of severe hypersensitivity reactions.

4.8 Undesirable effects

The frequency and severity of adverse events are generally similar between patients receiving Paclitaxel Seacross for treatment of ovarian, breast, or non-small cell lung carcinoma. These patients, if they also have AIDS-related Kaposi's sarcoma, may have more frequent and severe hematologic toxicity, infections (including opportunistic infections*), and febrile neutropenia. These patients require a lower dose intensity and supportive care. Elevated liver function tests and renal toxicity have a higher trend of incidence in KS patients as compared to patients with solid tumours.

*Opportunistic infections included cytomegalo virus, herpes simplex, Pneumocystis carinii, M. avium intracellulare, esophageal candidiasis, cryptosporidiosis, cryptococcal meningitis, and leukoencephalopathy.

The frequency of undesirable effects listed below is defined using the following convention:
very common; common; uncommon; rare; very rare; not known.
Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

Infections and infestations:	<i>Very common:</i> infection <i>Uncommon:</i> septic shock <i>Rare</i> *: pneumonia, sepsis
Blood and lymphatic system disorders:	<i>Very common:</i> myelosuppression, neutropenia, anaemia, thrombocytopenia, leucopenia, fever, bleeding <i>Rare:</i> febrile neutropenia <i>Very rare</i> *: acute myeloid leukaemia, myelodysplastic syndrome
Immune system disorders:	<i>Very common:</i> minor hypersensitivity reactions (mainly flushing and rash) <i>Uncommon:</i> significant hypersensitivity reactions requiring therapy (e.g., hypotension, angioneurotic oedema, respiratory distress, generalised urticaria, edema, back pain, chills) <i>Rare:</i> anaphylactic reactions (with fatal outcome) <i>Very rare</i> *: anaphylatic shock
Metabolism and nutrition disorders:	<i>Very rare</i> *: anorexia
Psychiatric disorders:	<i>Very rare</i> *: confusional stage
Nervous system disorders:	<i>Very common:</i> neurotoxicity (mainly: peripheral neuropathy), <i>Rare:</i> motor neuropathy (with resultant minor distal weakness) <i>Very rare</i> *: autonomic neuropathy (resulting in paralytic ileus and orthostatic hypotension), grand mal seizures, convulsions, encephalopathy, dizziness, headache, ataxia
Eye disorders:	<i>Very rare</i> *: reversible optic nerve and/or visual disturbances (scintillating scotomata), particularly in patients who have received higher doses than recommended, photopsia, vitreous floaters
Ear and labyrinth disorders:	<i>Very rare</i> *: ototoxicity, hearing loss, tinnitus, vertigo, ototoxicity
Cardiac disorders:	<i>Very common:</i> abnormal ECG <i>Common:</i> bradycardia <i>Uncommon:</i> cardiomyopathy, asymptomatic ventricular tachycardia, tachycardia with bigeminy, AV block and syncope, myocardial infarction <i>Very rare</i> *: atrial fibrillation, supraventricular tachycardia
Vascular disorders:	<i>Very common:</i> hypotension <i>Uncommon:</i> hypertension, thrombosis, thrombophlebitis <i>Very rare</i> *: shock
Respiratory, thoracic and mediastinal disorders:	<i>Rare</i> *: dyspnea, pleural effusion, respiratory failure, interstitial pneumonia, lung fibrosis, pulmonary embolism <i>Very rare</i> *: cough
Gastrointestinal disorders:	<i>Very common:</i> nausea, vomiting, diarrhoea, mucosal inflammation <i>Rare</i> *: bowel obstruction, bowel perforation, ischaemic colitis, pancreatitis <i>Very rare</i> *: mesenteric thrombosis, pseudomembranous colitis, oesophagitis, constipation, ascites
Hepato-biliary disorders:	<i>Very rare</i> *: hepatic necrosis(with fatal outcome), hepatic encephalopathy (with fatal outcome)
Skin and subcutaneous tissue disorders:	<i>Very common:</i> alopecia <i>Common:</i> transient and mild nail and skin changes <i>Rare</i> *: pruritus, rash, erythema, phlebitis, cellulitis, skin exfoliation, necrosis and fibrosis, radiation recall <i>Very rare</i> *: Stevens-Johnson syndrome, epidermal necrolysis, erythema multiforme, exfoliative dermatitis, urticaria, onycholysis (patients on therapy should wear sun protection on hands and feet)
Musculoskeletal, connective tissue disorders:	<i>Very common:</i> arthralgia, myalgia
General disorders and administration site conditions:	<i>Common:</i> injection site reactions (including localised oedema, pain, erythema, induration, on occasion extravasation can result in cellulitis) <i>Rare</i> *: asthenia, pyrexia, dehydration, oedema, malaise
Investigations:	<i>Common:</i> severe elevation in AST (SGOT), severe elevation in alkaline phosphatase <i>Uncommon:</i> severe elevation in bilirubin <i>Rare</i> *: increase in blood creatinine

Adverse event with combination treatment

Paclitaxel Seacross + Cisplatin

When administered as a three-hour infusion for the first-line chemotherapy of ovarian cancer, neurotoxicity, arthralgia/myalgia, and hypersensitivity were reported as more frequent and severe by patients treated with Paclitaxel Seacross followed by cisplatin than patients treated with cyclophosphamide followed by cisplatin. Myelosuppression appeared to be less frequent and severe with Paclitaxel Seacross as a three-hour infusion followed by cisplatin compared with cyclophosphamide followed by cisplatin.

When Paclitaxel Seacross is given in combinations with cisplatin 75 mg/m², the incidence of severe neurotoxicity is more common at a Paclitaxel Seacross dose of 175 mg/m² given by 3-hour infusion than at a dose of 135 mg/m² given by 24-hour infusion.

Patients treated with Paclitaxel Seacross and cisplatin may have an increased risk of renal failure during the combination therapy of paclitaxel and cisplatin in gynecological cancers as compared to cisplatin alone.

Paclitaxel Seacross + Trastuzumab

When Paclitaxel Seacross was administered as a 3-hour infusion in combination with trastuzumab for the first-line treatment of patients with metastatic breast cancer, the following events (regardless of relationship to Paclitaxel Seacross or trastuzumab) were reported more frequently than with single agent Paclitaxel Seacross: heart failure, infection, chills, fever, cough, rash, arthralgia, tachycardia, diarrhea, hypertension, epistaxis, acne, herpes simplex, accidental injury, insomnia, rhinitis, sinusitis, and injection site reaction. Some of these frequency differences may be due to the increased number and duration of treatments with Paclitaxel Seacross /trastuzumab combination vs single agent Paclitaxel Seacross. Severe events were reported at similar rates for Paclitaxel Seacross /trastuzumab and single agent Paclitaxel Seacross.

Administration of trastuzumab in combination with Paclitaxel Seacross in patients previously treated with anthracyclines resulted in an increased frequency and severity of cardiac dysfunction in comparison with patients treated with Paclitaxel Seacross single agent and rarely has been associated with death. In all but these rare cases, patients responded to appropriate medical treatment.

Paclitaxel Seacross + Doxorubicin

Congestive heart failure has been reported for combination therapy of Paclitaxel Seacross and doxorubicin in previously untreated patients with metastatic breast carcinoma and no prior chemotherapy. Cases of myocardial infarction have been reported rarely. Cardiac dysfunction and reduction of left ventricular ejection fraction or ventricular failure have been reported typically in patients on Paclitaxel Seacross who have received other chemotherapy, notable anthracyclines.

Paclitaxel Seacross + Radiotherapy

Radiation pneumonitis has been reported in patients receiving concurrent radiotherapy.

4.9 Overdose

There is no known antidote for Paclitaxel Seacross overdose. The primary anticipated complications of overdose would consist of bone marrow suppression, peripheral neurotoxicity and mucositis.

Overdoses in pediatric patients may be associated with acute ethanol toxicity.

5. Pharmacological properties

5.1 Pharmacodynamic properties

Mechanism of Action

Paclitaxel is a novel antimicrotubule agent that promotes the assembly of microtubules from tubulin dimers. It stabilizes microtubules by preventing depolymerization resulting

in the inhibition of the normal dynamic reorganization of the microtubule network essential for cellular functions. Paclitaxel also induces abnormal arrays or "bundles" of microtubules throughout the cell cycle and multiple asters of microtubules during mitosis.

5.2 Pharmacokinetic properties

The pharmacokinetics of paclitaxel have been evaluated over a wide range of doses, up to 300 mg/m², and infusion schedules, ranging from 3 to 24 hours and have been shown to be non-linear and saturable with a disproportionately large increase in Cmax and AUC with increasing dose accompanied by an apparent dose-related decrease in total body clearance.

Following intravenous administration, paclitaxel exhibits a biphasic decline in plasma concentrations. The initial rapid decline represents distribution to the peripheral compartment and elimination of the drug. The later phase is due, in part, to a relatively slow efflux of paclitaxel from the peripheral compartment. In patients treated with doses of 135 and 175 mg/m² given as 3- and 24-hour infusions, mean terminal half-life has ranged from 13.1 to 52.7 hours, and total body clearance has ranged from 12.2 to 23.8 L/h/m². Mean steady state volume of distribution has ranged from 198 to 688 L/m², indicating extensive extravascular distribution and/or tissue binding.

Variability in systemic paclitaxel exposure, as measured by AUC(0-~4) for successive treatment courses is minimal; there is no evidence of accumulation of paclitaxel with multiple treatment courses.

Distribution

On average, 89% of drug is bound to serum proteins; the presence of cimetidine, ranitidine, dexamethasone, or diphenhydramine does not affect protein binding of paclitaxel.

Metabolism

Paclitaxel was metabolized primarily to 6α-hydroxypaclitaxel by the cytochrome P450 isozyme CYP2C8; and to two minor metabolites, 3-β-hydroxypaclitaxel and 6α, 3'-p-dihydroxypaclitaxel by CYP3A4. The metabolism of paclitaxel to 6α-hydroxypaclitaxel was inhibited by a number of agents (see INTERACTION WITH OTHER MEDICINAL PRODUCTS AND OTHER FORMS OF INTERACTION section).

Elimination

After intravenous administration of 15-275 mg/m² doses of paclitaxel as 1-, 6-, or 24-hour infusions, mean values for cumulative urinary recovery of unchanged drug ranged from 1.3% to 12.6% of the dose. This indicates extensive non-renal clearance of paclitaxel. In patients administered a 225 or 250 mg/m² dose of radiolabeled paclitaxel as a 3-hour infusion, 14% of the radioactivity was recovered in the urine and 71% was excreted in the feces in 120 hours. Total recovery of radioactivity ranged from 56% to 101% of the dose. Paclitaxel represented a mean of 5% of the administered radioactivity recovered in the feces while metabolites, primarily 6α-hydroxypaclitaxel, accounted for the balance.

Special Populations

Renal impairment

The effect of renal impairment on the disposition of paclitaxel has not been investigated.

Hepatic impairment

The disposition and toxicity of paclitaxel 3-hour infusion were evaluated in patients with varying degrees of hepatic function. Relative to patients with normal bilirubin, plasma paclitaxel exposure in patients with abnormal serum bilirubin ≤2 times upper limit of normal (ULN) administered 175 mg/m² was increased, but with no apparent increase in the frequency or severity of toxicity. In patients with serum total bilirubin > 2 times ULN, there was a statistically nonsignificant higher incidence of severe myelosuppression, even at a reduced dose (110 mg/m²), but no observed increase in plasma exposure.

6. Pharmaceutical particulars

List of excipients

Ethanol anhydrous
Polyoxyl 35 castor oil (Macrogolglycerol ricinoleate 35)
Nitrogen

Incompatibilities

Polyoxyethylated castor oil can result in DEHP (di-(2-ethylhexyl) phthalate) leaching from plasticised polyvinyl chloride (PVC) containers, at levels which increase with time and concentration. Consequently, the preparation, storage and administration of diluted paclitaxel should be carried out using non-PVC-containing equipment. This medicinal product must not be mixed with other medicinal products except those mentioned in section *Preparation for Intravenous Administration*.

Special precautions for storage

Vial before opening:

Store below 30°C.

After opening before dilution:

Chemical

Artwork Panel	Artwork Ref: Paclitaxel-PIL-MA	Internal Code:	Artwork Code:
	Version No: 00	Date: 19/08-2023	Colour References: PMS1 K100 ■
Product: Paclitaxel	Pack Type: PIL	Substrate Weight: 60gsm	如果非汇宇场地包装 说明书印刷时需要确认是否折叠 HUIYU by Design
Client: Seacross Pharmaceuticals Limited	Dimensions: 240x480mm	Fonts: Arial,Arial-bold, Min Font Point: 7	